Unit 9, Lesson 8: Dilations on the Coordinate Plane

Benchmarks

MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.

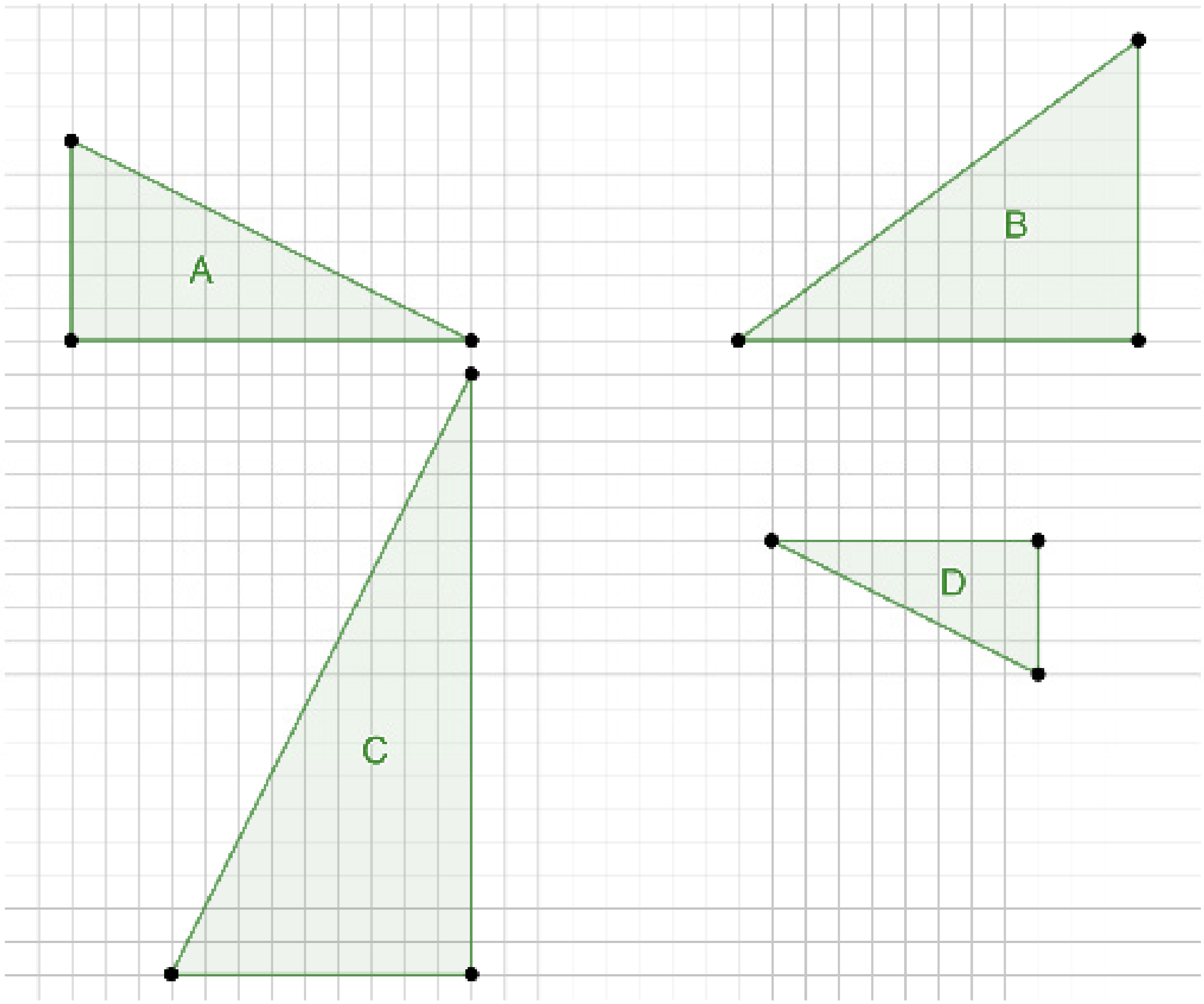
MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

Learning Targets

- ☐ **I can describe how an image and preimage generated by a single dilation are related.**
- ☐ **I can apply the effect of a dilation on the coordinate plane.**

9.8.1 Warm-Up: Which One Doesn't Belong?

1. Four triangles are shown.

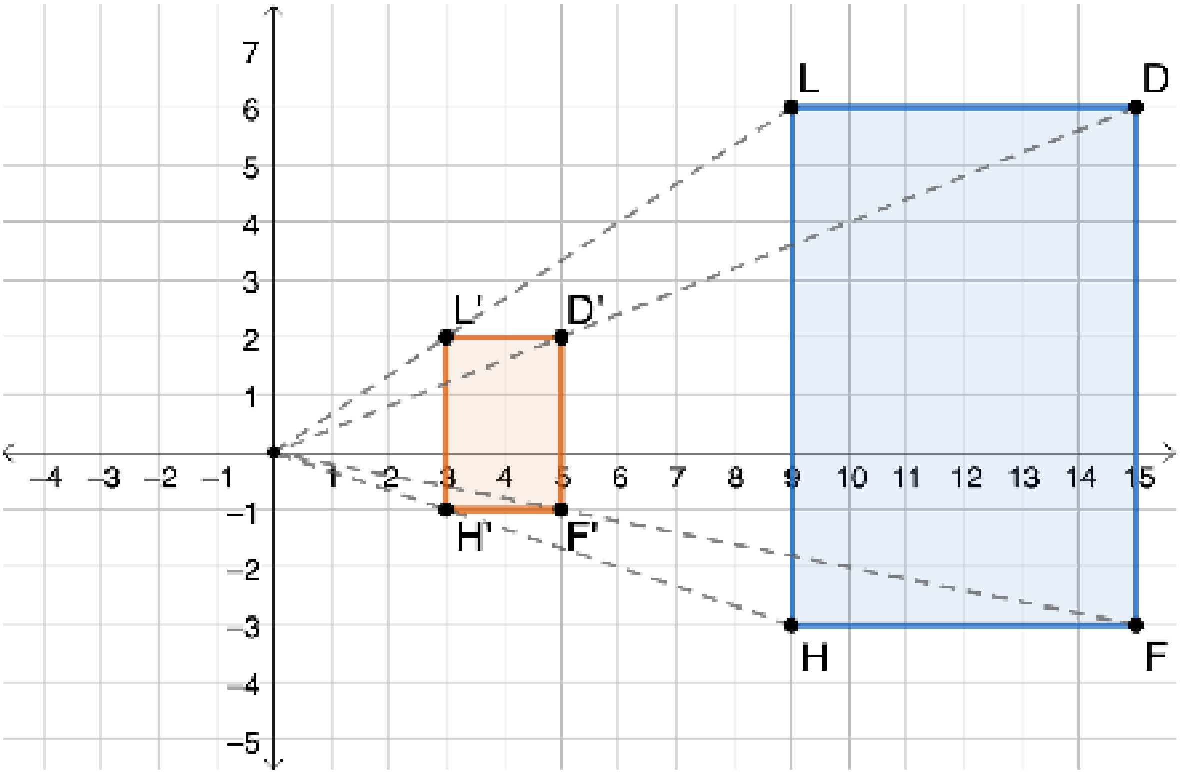


a. Which one doesn't belong?

b. Explain your reasoning.

9.8.2 Exploration: Dilations on the Coordinate Plane

1. On the coordinate plane, quadrilateral $DLHF$ was dilated with a scale factor of $\frac{1}{3}$ centered at the origin.



a. Complete the table.

Sides of the Preimage	Length	Sides of the Image	Length
Segment LD		Segment $L'D'$	
Segment DF		Segment $D'F'$	
Segment HF		Segment $H'F'$	
Segment LH		Segment $L'H'$	

b. Complete the statement.

The side lengths of quadrilateral $D'L'H'F'$ are

- $\frac{1}{3}$
- 3

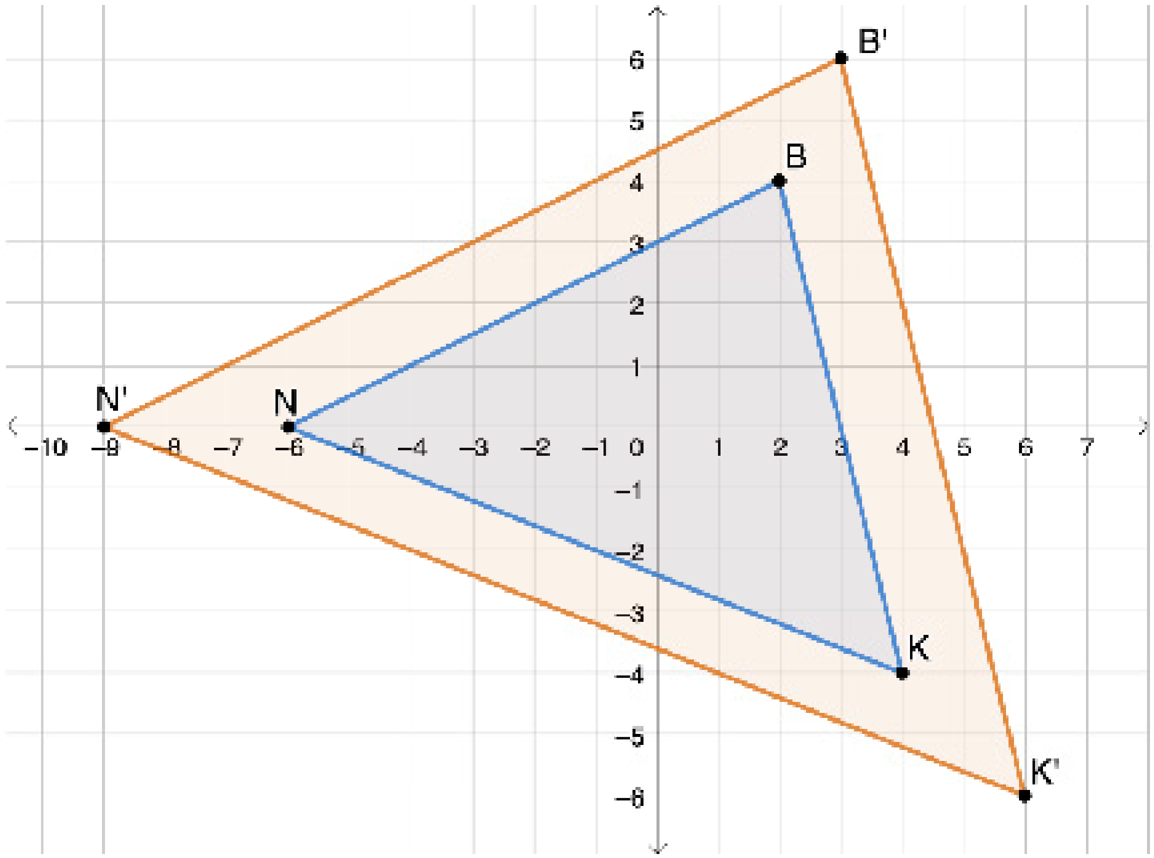
times the length of the sides of quadrilateral $DLHF$.

c. Complete the table with the coordinates of the vertices of the preimage and image.

Vertices of the Preimage	Coordinates	Vertices of the Image	Coordinates
Point D		Point D'	
Point L		Point L'	
Point H		Point H'	
Point F		Point F'	

d. What do you notice about the relationship between the coordinates of the corresponding points on the preimage and image?

2. On the coordinate plane, $\Delta B'K'N'$ is the result of a dilation of ΔBKN centered at the origin. The scale factor is 1.5.



a. Complete the table with the coordinates of the vertices of the preimage and image.

Vertices of the Preimage	Coordinates	Vertices of the Image	Coordinates
Point B		Point B'	
Point K		Point K'	
Point N		Point N'	

b. Discuss with your partner how the relationship between the coordinates is related to the scale factor. Summarize your discussion.

3. Suppose point W is the vertex of a polygon and has coordinates $(2, 10)$.

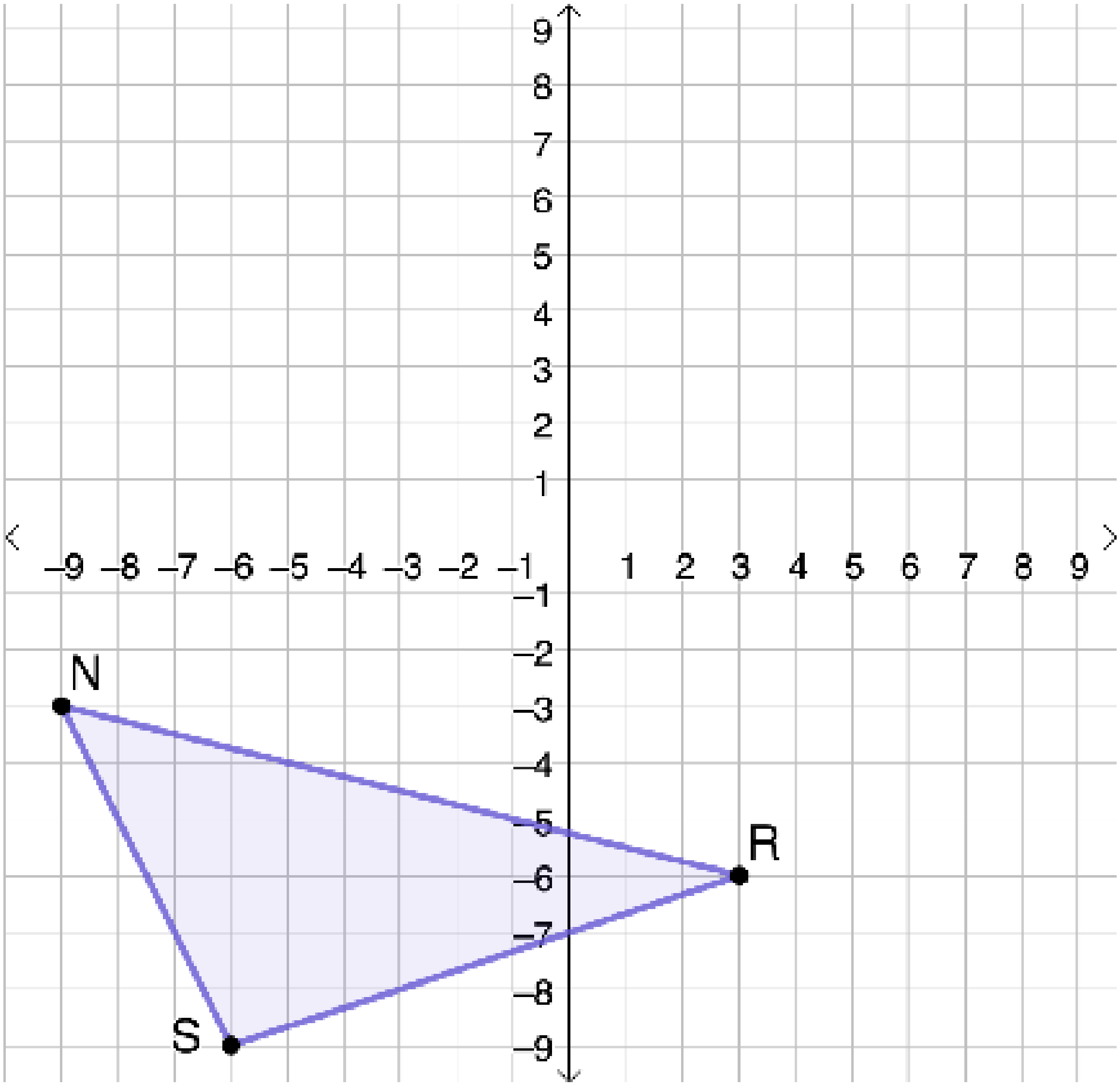
a. If this polygon were dilated by a scale factor of 2.7 centered at the origin, discuss with your partner how you could determine the coordinates of the image of point W.

b. Determine the coordinates of point W'.

9.8.3 Bring It Together: Dilations on the Coordinate Plane

When a dilation is performed on a figure on the coordinate plane with its center at the origin, the coordinates of the image are equal to the coordinates of the preimage multiplied by the scale factor.

1. Triangle NRS is plotted on the coordinate plane.



a. Describe how the image of the dilation of ΔNRS, $\Delta N'R'S'$, compares to the preimage, ΔNRS, using a scale factor of $\frac{2}{3}$ centered at the origin.

b. Complete the table.

Vertices of the Preimage	Coordinates	Vertices of the Image	Coordinates
Point N		Point N'	
Point R		Point R'	
Point S		Point S'	

c. Draw and label $\Delta N'R'S'$.

d. Use patty paper to verify corresponding angles on the preimage and image have the same measure.

e. Complete the statements.

Triangle NRS and triangle $N'R'S'$ are
- o congruent
- o similar

triangles. The side lengths of
- o ΔNRS
- o $\Delta N'R'S'$

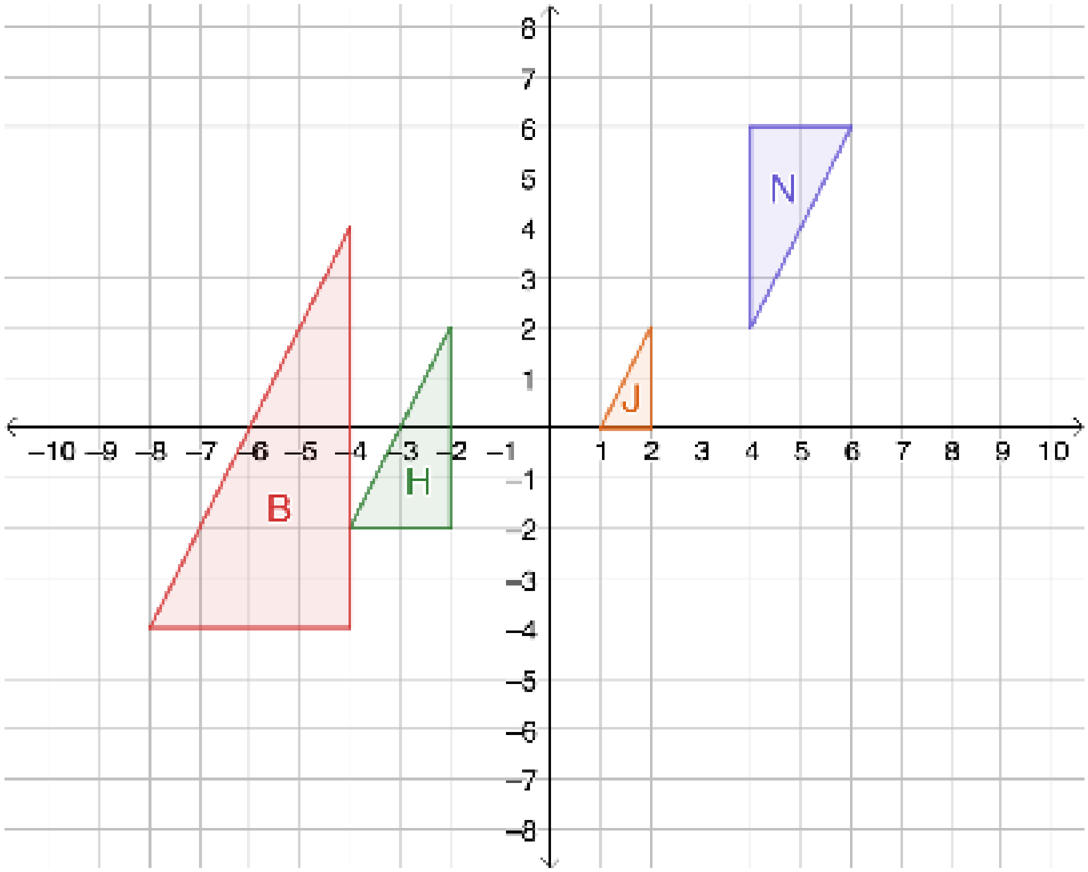
are $\frac{2}{3}$ times the side lengths of
- o ΔNRS.
- o $\Delta N'R'S'$.

The corresponding angles of the triangles are
- o congruent.
- o similar.

9.8.4 Guided Practice: Describe the Relationship

Work with your partner to complete the following.

1. Triangles *B*, *H*, *J*, and *N* are figures on the coordinate plane.



a. Discuss with your partner which triangles are similar or congruent. Then, complete the table.

Triangle Pair	Are they similar?	Are they congruent?
J and *N*	o Yes o No	o Yes o No
B and *H*	o Yes o No	o Yes o No
H and *N*	o Yes o No	o Yes o No
J and *B*	o Yes o No	o Yes o No

b. **Complete the statement.**

Triangle ____ is the image of triangle ____ after a dilation using a scale factor of $\frac{1}{2}$ centered at the origin.

c. **Explain how you determined your answer to Part B.**

d. **Work with your partner to describe at least three characteristics of the relationship between the triangles selected in Part B.**

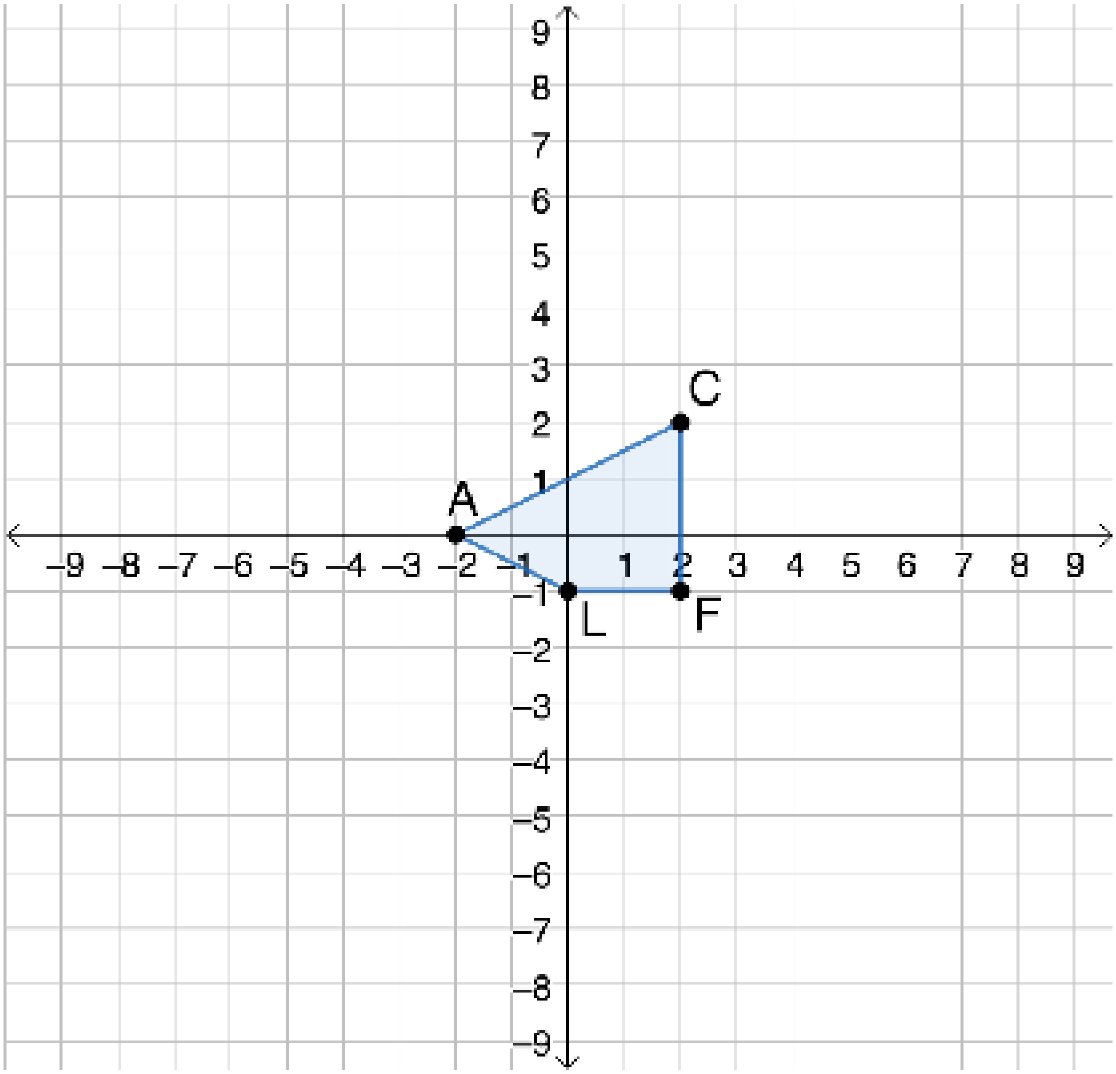
1.

2.

3.

9.8.5 Your Turn!

1. Quadrilateral *ACFL* is plotted on the coordinate grid.



a. If the figure were dilated by a scale factor of 4 with the origin as the center of dilation, what are the coordinates of the vertices *A′C′F′L′*?

Vertex	Coordinates
A′	
C′	
F′	
L′	

b. Dilate quadrilateral $ACFL$ using a scale factor of 4 centered at the origin. Label the image $A'C'F'L'$.

c. Explain whether the figures are congruent, similar, or neither.

9.8.6 Wrap-Up: Growth Mindset

Complete each statement based on today's lesson.

1. **Today I am still unsure about …**

2. **I will try to build my confidence by …**